

Kaustav Dey

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PROFILE

AI/ML Engineer experienced in building **production-grade AI systems** including **fraud detection engines, RAG pipelines, document intelligence systems, and multi-agent AI workflows**. Skilled in **Agentic AI, LLMs, RAG, Azure/AWS-based AI infrastructure, NLP**, and scalable ML systems using **Python, LangChain, CrewAI, Redis, and PostgreSQL**.

EDUCATION

Techno Main Salt Lake

Kolkata, India

B.Tech in Computer Science Engineering, achieved a **CGPA of 8.52**

Jul 2019 – Jul 2023

EXPERIENCE

NexDev IT Solutions

Kolkata, India

AI/ML Engineer

Jan 2026 – Present

- Architected a production-grade **real-time Fraud Detection System** using hybrid **Rule Engine + XGBoost** pipelines, achieving **96.2% ROC-AUC and 90.5% fraud recall** across **111K+ e-commerce transactions**.
- Engineered scalable ML infrastructure with **Redis-backed feature stores, PostgreSQL** fallback pipelines, and **gRPC microservices** for fault-tolerant **real-time inference**.
- Built a topology-aware **Resume Content Extraction System** without LLM dependency using deterministic NLP and layout reconstruction algorithms, **reducing inference cost** while accurately parsing complex multi-column resumes.

NoBroker

Bangalore, India

Data Research Intern (Data Science & Insights)

Nov 2024 – May 2025

- Conducted competitor research, analyzed data, ensured data validity, consistency, and reliability, and delivered business insights.
- Developed a **Python web scraper** to **automate** competitor data collection, boosting data collection **productivity by 80%** and supporting the SEO team's marketing efforts.

Intellipaath Software Solutions Pvt. Ltd.

Bangalore, India

Data Science Intern

Aug 2023 – Nov 2024

- Developed **Decision Trees & Random Forests** for fraud detection, boosting accuracy by **30%** and reducing false positives by **25%**.
- Developed **deep learning models** for telecom churn prediction, achieving **85% accuracy** and improving customer retention by **20%**.

PROJECTS

RAG-Powered Document Q&A App (Groq API and Llama 3.1)

Sep 2025 – Sep 2025

- Engineered a lightweight **RAG System** using **LangChain, FAISS, and Groq Llama 3.1**, enabling context-aware Q&A chatbot.
- Optimized **RAG pipeline** with efficient **text chunking** and **vector retrieval**, reducing response time by **30%** and boosting retrieval precision on benchmark queries.

Multi-Agent AI Research Analyst System (LangChain, CrewAI and Groq)

Oct 2025 – Oct 2025

- Developed an **AI-powered research automation system** using **CrewAI and LangChain**, orchestrating 4 specialized agents (**Researcher, Summarizer, Strategist, Presenter**) for end-to-end market intelligence.
- Integrated **Groq Llama 3.1** and **Serper.dev API** for real-time research, automated summarization, and **SWOT analysis**.

SKILLS

Technical: Python, SQL, Machine Learning, Deep Learning, NLP, Agentic AI, Generative AI, RAG, Multi-Agent Systems, LLM Integration, LangChain, CrewAI, Hugging Face, Scikit-Learn, TensorFlow, PyTorch, Vector Databases, GraphRAG, Redis, Azure MLOps, AWS.

Tools: Power BI, Microsoft SQL Server, Azure MLOps, Jira, Figma, Jupyter Notebook, Visual Studio Code, Docker.

Soft Skills: Stakeholder Requirement Analysis, Project Management, Problem-Solving, Leadership, Analytical, Critical Thinker.

CERTIFICATIONS

- [Advanced Certification in Data Science and AI conducted by iHub DivyaSampark at IIT Roorkee](#)
- [AI Agents Intensive Course with Google](#)
- [Microsoft Excel Advanced Excel Formulas & Functions](#)